

CAN Command: Temperature Telemetry

Purpose:

Periodically report motor and controller temperatures to the master for monitoring and logging.

CAN Frame Specification:

CAN ID: 0x2D0 + NodeID (e.g., 0x2D2 for front driver, 0x2D4 for rear driver)

Direction: Motor Driver * Master

Payload:

Byte	Field	Type	Description
0-1	MotorTempLeft	int16	Left motor temperature (°C × 10)
2-3	MotorTempRight	int16	Right motor temperature (°C × 10)
4	MosfetTempLeft	int8	Left MOSFET temperature (°C)
5	MosfetTempRight	int8	Right MOSFET temperature (°C)
6	CpuTemp	int8	Controller CPU temperature (°C)
7	Reserved	uint8	Reserved for future use

Action:

Sent periodically (e.g., every 100 ms) or on master request.

Notes:

All values are instantaneous readings.

CAN Command: Report Odometer and Average Current

Purpose:

Periodically report total odometer (mileage) and average current to the master.

CAN Frame Specification:

CAN ID:

(e.g.,
for front,
for rear)

Direction: Motor Driver * Master

Payload: {#payload-2}

Byte	Field	Type	Description
0-3	U32Global_OdoMilage	uint32	Total odometer, meters
4-5	U16Global_TripMilage	uint16	Trip odometer, meters
6-7	S32Global_AvrCurrentConvert	int16	Average phase current, mA

Action:

Sent periodically every 1000 ms or on master request.

CAN Command: System History Read Error

CAN ID:

(e.g.,
for front,
for rear)

Direction: Motor Driver * Master

Payload:

Byte	Field	Type	Description
0	fGlobal_ReadSystemHistoryInformationErr	uint8	1 Error, 0 OK

CAN Command: DriveCommand

Purpose:

Unified command to control both left and right motors with a single message, including throttle and mode.

CAN Frame Specification:

CAN ID: 0x300 + NodeID (e.g., 0x302 for front, 0x304 for rear) **Direction:** Master → Motor Driver

Payload:

Byte	Field	Type	Description
0-1	ThrottleLeft	int16	-1000 (max reverse) to +1000 (max forward), 0=no power
2-3	ThrottleRight	int16	-1000 (max reverse) to +1000 (max forward), 0=no power
4	MotorMode	uint8	Bits 0–1: Motor mode (see table below); Bit 2: DisableBraking (see table below); Bits 3–7: Reserved (set to 0)
5-6	Limit	uint16	If MotorMode is Speed, Limit = MaxTorque (nM). If MotorMode is Torque, Limit = MaxSpeed (RPM).
7	Reserved	—	Reserved for future use

MotorMode Bit Allocation (Byte 4)

Bits	Value	Name	Description
0–1	00	No mode	No mode selected
0–1	01	Speed	Speed control mode
0–1	10	Torque	Torque control mode
0–1	11	—	Reserved

2	0	—	Braking enabled
2	1	DisableBraking	Braking disabled
3–7	—	Reserved	Set to 0

Note:

- The meaning of the Limit field depends on the MotorMode:
 - If MotorMode is Speed (01): Limit sets the maximum allowed torque (in nM).
 - If MotorMode is Torque (10): Limit sets the maximum allowed speed (in RPM).
- Every reserved field must be set to 0.
- For DisableBraking (bit 2): 0 = Braking disabled, 1 = Braking enabled.
- When receiving DisableBraking in the ExtendedPower command, the driver must release the regenerative brake.

CAN Command: SpeedTelemetry

Purpose:

Unified telemetry response for speed and torque.

CAN Frame Specification:

CAN ID: 0x310 + NodeID (e.g., 0x312 for front, 0x314 for rear) **Direction:** Motor Driver → Master

Payload:

Byte	Field	Type	Description
0-1	SpeedLeft	int16	Left motor speed (RPM)
2-3	SpeedRight	int16	Right motor speed (RPM)
4-5	TorqueLeft	int16	Left motor torque (nM)
6-7	TorqueRight	int16	Right motor torque (nM)

CAN Command: MotorTelemetry

Purpose:

Report power usage statistics for the system and each motor.

CAN Frame Specification:

CAN ID: 0x320 + NodeID (e.g., 0x322 for front, 0x324 for rear) **Direction:** Motor Driver → Master

Payload:

Byte	Field	Type	Description
0	TotalAmps	int8	Total current ($A \times 1$)
1-2	TotalVoltage	uint16	Total voltage ($V \times 10$, 0.1V precision)
3	RightMotorAmps	int8	Right motor current ($A \times 1$)
4	LeftMotorAmps	int8	Left motor current ($A \times 1$)
5	FaultLevel	uint8	Fault level/status code (see table below)
6	MotorMode	uint8	Bits 0–1: Motor mode (see table below); Bit 2: DisableBraking (see table below); Bits 3–7: Reserved (set to 0)
7	Reserved	—	Reserved for future use

MotorMode Bit Allocation (Byte 6)

Bits	Value	Name	Description
0–1	00	No mode	No mode selected
0–1	01	Speed	Speed control mode

0–1	10	Torque	Torque control mode
0–1	11	—	Reserved
2	0	—	Braking enabled
2	1	DisableBraking	Braking disabled
3	0	—	Left motor is Not braking
3	1	brakingMotorLeft	Left motor regenerative braking active
4	0	—	Left motor not standing
4	1	StandingMotorLeft	Left motor RPM < MIN_RPM_STANDING
5	0	—	Right motor not braking
5	1	BrakingMotorRight	Right motor regenerative braking active
6	0	—	Right motor not standing
6	1	StandingMotorRight	Right motor RPM < MIN_RPM_STANDING
7	—	Reserved	Set to 0

FaultLevel Values (Byte 5)

Value	Meaning
0	None (no fault)
1	Warning (starting to be hot)
2	Degraded (performance reduced, e.g., due to "MotorStartProtectT")
3	Stopping (initiating stop due to "MotorEndProtectT")
4	Stopped (fully stopped due to "MotorEndProtectT" or other critical condition)
5-255	Reserved

Note:

- Every reserved field must be set to 0.
- For DisableBraking (bit 2): 0 = enabled, 1 = disabled.
- For brakingMotorLeft (bit 3): 0 = Left motor not braking, 1 = Left motor regenerative braking active.
- For StandingMotorLeft (bit 4): 1 = Left motor RPM is less than MIN_RPM_STANDING (as set in code).
- For StandingMotorRight (bit 5): 1 = Right motor RPM is less than MIN_RPM_STANDING (as set in code).
- For brakingMotorRight (bit 6): 0 = Right motor not braking, 1 = Right motor regenerative braking active.

CAN Command: CurrentLimitCmd

Description:

Sets the maximum discharge and charge current limits for the motor driver.

Direction:

Master → Motor Driver

CAN Frame Specification:

CAN ID: 0x400 + NodeID (e.g., 0x412 for front, 0x414 for rear) **Direction:** Master → Motor Driver **Sent:** On command or periodically (e.g., every 100 ms) if limits are being actively managed

Payload:

Byte	Field	Type	Description
0-1	DCL	uint16	Discharge current limit (mA): max current the driver can draw from the batteries. Range: 0...65535 mA
2-3	CCL	uint16	Charge current limit (mA): max current the driver can push to the batteries. Range: 0...65535 mA
4	Reserved	—	Reserved for future use
5	Reserved	—	Reserved for future use
6	Reserved	—	Reserved for future use
7	Reserved	—	Reserved for future use

CAN Command: AlarmsFaultsRpt

Description:

Periodically reports active alarms and fault codes to the master.

Direction:

Motor Driver → Master

CAN Frame Specification:

CAN ID: 0x500 + NodeID (e.g., 0x512 for front, 0x514 for rear) **Direction:** Motor Driver → Master **Sent:** Periodically every 100 ms

Payload:

Byte	Field	Type	Description
0-1	CpuFaultCode	uint16	Highest CPU fault code (or bitmask of CPU faults).
2-3	MotorLeftFaultCode	uint16	Highest left motor fault code (or bitmask).
4-5	MotorRightFaultCode	uint16	Highest right motor fault code (or bitmask).
6	Reserved	—	Reserved for future use.
7	Reserved	—	Reserved for future use.

Note:

- Fault code fields may be used as either a single code or a bitmask, depending on system design.
- Reserved bytes must be set to 0.